

ZIHENG QIU

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EDUCATION

Fudan University, Ph.D. in Computer Science and Technology Sep. 2025 - Present

- Research: embodied AI and robot manipulation; Advisor: Prof. Wenchao Ding

University of Electronic Science and Technology of China, B.Eng. in Software Engineering Sep. 2021 - Jun. 2025

- Advisor: Prof. Yong Liao

PUBLICATIONS

AutoSpeed: Annotation-Free Stage-Adaptive Motion Speed Learning for Robot Manipulation *ECCV 2026*

Qingda Hu[†], **Ziheng Qiu**[†], *et al.* († Equal contribution)

[arXiv](#) [Project](#)

- Proposed a general learning framework that learns stage-specific execution speeds end to end without speed annotations, improving policy performance beyond the training data.
- Validated the method in Aloha, Meta-World, and LIBERO simulations and on an Agilex Piper bimanual robot, achieving a $1.78\times$ average real-world speedup while improving task success.

Learning Temporally Contextualized Latent Action Space for Robot Manipulation *ICLR 2027*

Qingda Hu[†], **Ziheng Qiu**[†], *et al.* († Equal contribution; under review)

- Proposed a temporally contextualized latent action space that organizes action trajectories into continuous, compact representations with task-stage information, enabling learning in a more structured prediction space.
- Improved policy success rates, training convergence, and action smoothness in RoboTwin and real-robot experiments.

Resolving State Ambiguity in Robot Manipulation via Adaptive Working Memory *RA-L 2026*

Recoding

Qingda Hu, **Ziheng Qiu**, *et al.*

[arXiv](#) [Project](#)

- Proposed PAM, a visuomotor policy with adaptive working memory that uses hierarchical feature extraction to encode actions and context, compressing multi-scale history into compact memory to resolve state ambiguity.
- Trained a single policy across seven real-world state-ambiguity tasks with a history window of up to 300 frames (about 10 seconds), achieving an average success rate of 0.91.

HONORS

Graduate Academic Scholarship, Fudan University 2026

SKILLS

- **Programming and deep learning:** Python and PyTorch; model training, debugging, evaluation, and experimental analysis; familiar with Linux, Bash, Git, and Docker
- **Generative and multimodal models:** diffusion models, large language models, and vision-language models for visuomotor policies and robot manipulation
- **Robot learning:** imitation learning, reinforcement learning, action representations, world models, and long-horizon modeling
- **Simulation and deployment:** LIBERO, Meta-World, and RoboTwin; deployment and debugging on Franka and AgileX single- and dual-arm platforms